



Review

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Recent advances in design, sensing and control have led to aerial robots that offer great promise in a range of real-world applications. However, one critical open question centres on how to improve the energetic efficiency of aerial robots so that they can be useful in practical situations. This review paper provides a survey on small-scale aerial robots (i.e. less than 1 m² area foot print, and less than 3 kg weight) from the point of view of energetics. The paper discusses methods to improve the efficiency of aerial vehicles, and reports on recent findings by the authors and other groups on modelling the impact of aerodynamics for the purpose of building energy-aware motion planners and controllers.

1. Introduction

Small-scale unmanned aerial vehicles (UAVs) offer promise in several domains that span from core academic research to defence and commercial applications. Some key examples include robotic first responders [1], aerial manipulation [2], cooperative construction [3], radiation detection [4] and intelligence, surveillance and reconnaissance. Bioinspired aerial robots [5,6] can help study the trade-offs in avian and insect flight. To realize the full potential of aerial vehicles in applications, it is important to further insist on design and development.

The design and development process in small aerial robots has three steps: building autonomy, improving reliability and testing across complex environments. The process is outlined in figure 1. Building autonomy ties to developing and unifying algorithms for localization and mapping [7], state estimation [8], motion planning [9,10], robotic vision [11] and control [12,13]. The developed solutions are typically tested in (semi-)controlled environments. Once a system is capable of autonomous operation, improving its reliability follows. This step is related to optimizing the system design according to practical operational needs. Such needs include appropriate integration of software and hardware, endurance optimization, and improvement of payload capacity and system energetics. As in the previous step, developed solutions are typically tested in (semi-)controlled settings. What follows then is extensive testing by researchers, practitioners and users, in a wide variety of applications and environments. Testing in practical situations, however, may in fact reveal limitations in the reliability and autonomy of the aerial system at hand, leading the designer back to one of the two previous steps (figure 1). These steps apply, in principle, to the design and development of any aerial robot configuration.

1.1. Small-scale unmanned aerial vehicle configurations

Several UAV configurations at small scales have been studied (figure 2). The most common configurations include fixed-wing aircraft, rotary-wing vehicles and bioinspired flapping-wing designs [14]. Fixed-wing aircraft were introduced first, drawing from the design methods for larger vehicles and utilizing appropriate scaling laws [15]. They require low thrust-to-weight ratio to operate [16] because wings provide lift, and they typically fly at high speeds. They are also more energy-efficient in forward flight than rotary-wing vehicles [17]. However, their inability to hover, or fly at low speeds when needed, makes them less suitable than rotary-wing vehicles for manoeuvring in cluttered environments [18].

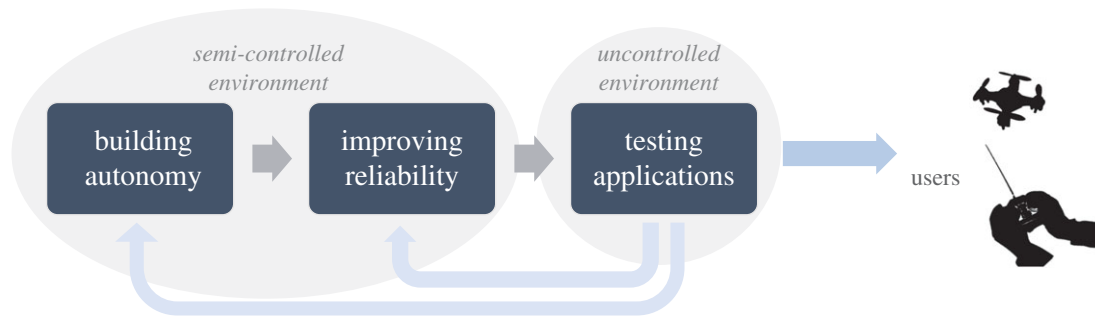


Figure 1. Conceptual design and development process in small-scale aerial robots. The graph essentially summarizes the key steps to successfully steer autonomous aerial robots from the research and development phase to practical applications. (Online version in colour.)

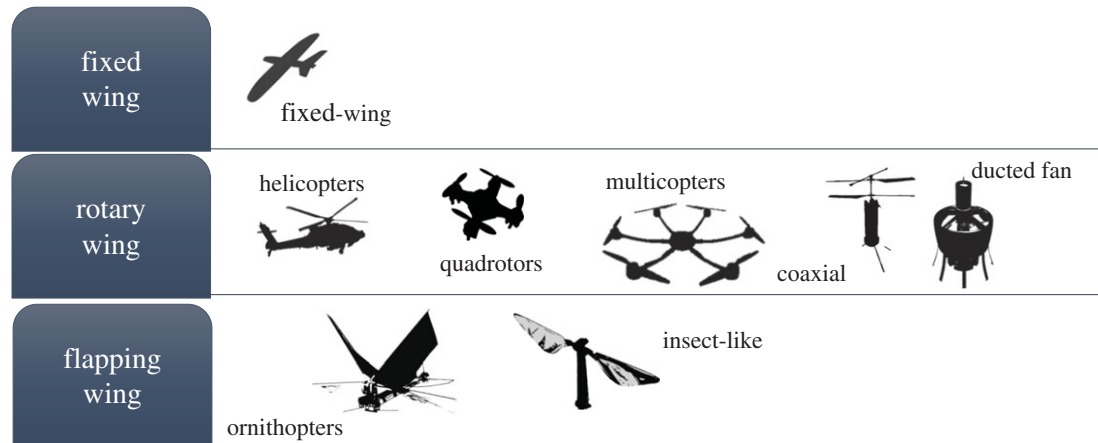


Figure 2. Illustrations of various UAV configurations. Each type of UAV has its own strengths and weaknesses, and has received different degree of adoption in applications. (Online version in colour.)

Rotary-wing vehicles can execute vertical take-off and landing (VTOL), hover and are highly manoeuvrable. These features make them exceptional candidates for operation in confined spaces [18,19]. The design of rotorcraft vehicles is an active area of research, and several approaches have been proposed. Successful designs include traditional helicopters, quadrotors and vehicles with more than four rotors (multicopters), as well as coaxial [20–22] and ducted-fan [23] rotorcraft. Among these designs, the quadrotor stands out due to its simplicity in terms of fabrication and control. Furthermore, the quadrotor offers promise in many applications [18], and it has a low cost of entry. As a result, it has been widely adopted in both academia and industry.

In several cases (e.g. going through crevices or in swarms), there is need to shrink the aerial vehicles down to the sub-millimetre scale [24]. However, fixed-wing and rotary-wing vehicles do not scale down well [17,25]. The miniaturization problem arises primarily because the propulsion systems employed in these vehicles perform poorly when scaled down [25]. This inefficiency is further aggravated by the unsteady aerodynamics that govern flight at low Reynolds numbers [26]. An alternative paradigm to circumvent these challenges is the introduction of bioinspired flapping-wing robots [27,28].

Flapping-wing vehicles include bird-like ‘ornithopters’ [5,29–32] and insect-like robots [6,33,34]. Ornithopters generate lift by flapping their wings with synchronized small variations of angle of incidence. Their wings also contribute to thrust generation for forward flight. Ornithopters generally lack VTOL capabilities (with a few exceptions [30]), and they need to obtain an initial airspeed to take off [35,36]. They are more agile than fixed-wing vehicles [27], but aeroelasticity and the fluid–structure coupling are currently not well understood [37]. Insect-like aerial robots rapidly change the

angle of incidence to generate lift and thrust. They are capable of VTOL, hovering and forward flight [6,33]. However, their advantage over rotary-wing vehicles in terms of efficiency [25], endurance or manoeuvrability is not yet clear. Furthermore, they increase the mechanical and controls complexity [38]. Despite recent progress in navigation and control [39–42], autonomous operation of flapping-wing vehicles appears to remain limited.

Table 1 outlines how four types of aerial vehicles that are currently capable of autonomous operation compare to each other. The comparisons are made on the basis of six important criteria for operation in practical situations. These are manoeuvrability, control simplicity, endurance, payload capacity, mechanical simplicity and low cost of entry. From table 1, it can be readily verified that vehicles capable of hovering are currently best suited for applications that require autonomy. Unfortunately, VTOL and hovering have high power requirements [18,43], rendering the improvement of the energetics of robotic flight even more pressing.

Remark: Most of the results presented herein are obtained from studies with quadrotors. The quadrotor has been proven successful in a variety of contexts [18], and thus offers a trustworthy testbed for studies in autonomy and energetics in particular. Nonetheless, the ideas put forward in the reported results can also be generalized to other small-scale UAV configurations.

1.2. Energetics of small rotorcraft

Despite the significant advances made in the area of aerial robots [17,18], one remaining key challenge centres on sustaining long-term autonomous operation. Unfortunately, current maximum flight times under ideal operational conditions are

Table 1. Assessment of key characteristic features for aerial vehicles used in applications that require autonomy.

vehicle type	fixed-wing	helicopter	quadrotor	multicopter
manoeuvrability in tight spaces	++	+++	++++	+++
control simplicity	+	++	++++	++++
endurance	++++	+++	++	++
payload capacity	++	+++	+++	++++
mechanical simplicity	+++	+	++++	+++
low cost of entry point	++	+++	++++	+++

restrictive for most practical applications. Maximum flight times for centimetre-scale aerial vehicles (e.g. the Crazyflie (<https://www.bitcraze.io/crazyflie-2/>) and Pico [44] quadrotors) are approximately 5–7 min. Larger, decimetre-scale vehicles such as Asctec's Pelican (<http://www.asctec.de/en/uav-uas-drones-rpas-roav/asctec-pelican/>) and DJI's Phantom 4 (<http://www.dji.com/phantom-4>) quadrotors can operate for up to 30 min. In typical operational conditions, the expected flight times for decimetre-scale quadrotors range between 10 and 20 min [18], depending on payload and aggressiveness of flight. In this light, how can we improve the capacity of aerial robots for reliable, long-term operation?

To answer this question one can look at (i) improving the efficiency of batteries or other power sources, (ii) developing automated recharging methods, and (iii) improving the vehicles' energy efficiency. The efficiency of power sources is typically captured well by the source specific energy and power [43]. The source specific energy (measured in Wh kg^{-1}) represents the total energy being available to the system. The source specific power (measured in W kg^{-1}) relates to the power that can be delivered to the vehicle instantaneously. To date, most aerial robots employ lithium–polymer (LiPo) batteries [18]. Such batteries offer acceptable specific energy at high specific power (typically in the range $100\text{--}150 \text{ Wh kg}^{-1}$ at over 1000 W kg^{-1}), and retain performance over hundreds of charging/discharging cycles [45]. Also, many different battery sizes to select from according to the application at hand are commercially available. At the same time, quadrotors have been found to consume about 200 W kg^{-1} on average [18,43]. To improve endurance, there is need for power sources with higher specific energy instead. To this end, a promising alternative would be lithium–sulfur (LiS) batteries. Such batteries can store double or more specific energy than LiPo batteries at similar or even higher specific power levels [46]. However, they are not commercially available yet. Other power options for which prototype UAVs exist include (hydrogen) fuel cells for rotorcraft (<http://www.hus.sg/hydrogen-multi-rotor/>) [47], and micro-rockets [48] for insect-like vehicles. Another interesting power source would be micro thermo-photo-voltaic (μTPV) cells [49,50], but scaling important components for system integration is still under research.

Some applications (e.g. persistent monitoring or airborne delivery tasks) require redundancy in terms of the number of vehicles to be deployed. In such cases, a different approach would be to use docking stations for automatic battery charging and/or swapping. Charging-only docking stations typically feature contact-based battery charging [51], or charging using a tether.¹ As an alternative, one can consider using devices and methods to both swap and charge batteries

[53–55]. Such approaches can reduce the time a vehicle remains idle and thus increase efficiency and vehicle utilization [55]. Despite these benefits, using docking stations adds another layer of complexity, which may not always be desirable. In some cases (e.g. availability of direct line of sight, and operation in non-cluttered environments), remote charging through laser [56] or magnetic resonance [57] power beaming could be another possible solution.

A third approach to boost the long-term operation of UAVs is by designing systems that consume as little energy as possible, and are energy-efficient. The main purpose of this paper is to identify ways to improve the energy efficiency of aerial robots, and discuss relevant open challenges.

1.3. How to improve energy efficiency?

To address the energy consumption challenge, this review paper focuses on recent advances in hardware and algorithm optimization, and multimodal locomotion.

1.3.1. Hardware-based optimization for energy efficiency

Perhaps the most straightforward way to reduce energy consumption is to minimize the weight of the vehicle. This can be achieved by using light-weight manufacturing materials [58], electronics² and sensors (such as cameras). In addition, careful component selection and system design [59,60] can help further. Hardware-based optimization is in fact well studied; here we present some of the most promising results.

1.3.2. Algorithm-based optimization for energy efficiency

Less studied but equally significant is the development of novel motion planning and control algorithms that are energy-aware. Energy-aware algorithms reduce energy consumption and extend flight times through the design of energy-efficient trajectories to be tracked by an aerial vehicle [61–64]. These algorithms can also benefit by using models that explicitly take into consideration various aerodynamics that affect energy consumption. To this end, we present results that seek to model rapid descent [65,66], drag [65,66] and ground effects [65,67–69]. We also present our preliminary new results that relate energy consumption to high-speed flight, indoors. Taken together with weight optimization, software-based solutions to save energy can lead to aerial robots capable of sustaining long-term operation.

1.3.3. Bio-inspired multimodal operation capabilities

More traditional works in design, planning and control for robotic flight focus on monolithic, flying-only approaches. This is in stark contrast with nature, where multimodal

behaviours are actually the norm (e.g. flying and walking). Drawing inspiration from nature, the focus on robotic flight has started to shift toward explicitly considering multimodal robot design and operation [17]. Doing so may have significant impact toward better energy utilization depending on the given task. We present results on aerial vehicles that can perch [70–72], walk [73,74], run [35] and roll [75].

Overall, improving the capacity of aerial robots for reliable and persistent operation will be valuable in several ways. Foremost, it will solidify the practical utility of aerial robots in real-world applications. Second, long-term operation will enable research on developing autonomous motion planning, control and machine learning algorithms that require long-term operation data. Furthermore, it will allow the experimental validation of various deployment [1,76] and persistent monitoring [77,78] algorithms.

1.4. Organization

The remainder of this review paper is organized as follows. Section 2 focuses on hardware-based optimization to reduce weight. Section 3 expands on algorithm-based approaches to improve energy efficiency, and reports on our preliminary new results on energetics of high-speed flights. Section 4 presents recent efforts on designing multimodal aerial robots. We discuss main findings at the end of each of §§2–4, and conclude in §5. For the convenience of the reader, we provide a list of symbols used in this review in appendix A.

2. Hardware-based efficiency optimization

This section focuses on how we can design energy-efficient systems from a hardware point of view. Reducing unnecessary weight either by careful component selection or system design will make a big impact to the energy consumption of a vehicle at hand.

2.1. Weight reduction

A direct way to reduce weight is to optimize the airframe design of the aerial vehicle. One example is to use airframes made of carbon fibre. Owing to its low weight and high strength properties [58], carbon fibre has already become the material of choice for large aircraft design [79]. It has also found use in aerial robots across scales. Examples range from the insect-scale robobee [6] to the popular 540 mm Asctec Hummingbird quadrotor (<http://www.asctec.de/en/uav-uas-drones-rpas-roav/asctec-hummingbird/>) and the 2 m wingspan ornithopter Phoenix [29]. Another way to further reduce the weight of an aerial robot is to design its printed circuit board (made of fibreglass [80]) so that it can serve as the main part of the airframe. While fibreglass may be a non-optimal material choice for airframe design [58,79], it offers an elegant way to scaling quadrotors down to the centimetre scale [44].

Furthermore, advances in manufacturing electronic components can provide an alternative leeway to reducing weight. Indeed, we witness a period at which improved computing and sensing packages for robotics applications are more available than ever. For example, in our experiments reported in §3.3 we use a 16 g Odroid-c0 microcomputer (http://www.hardkernel.com/main/products/prdt_info.php) on an Asctec Hummingbird quadrotor. Despite its size, this microcomputer is capable of running onboard all control algorithms for

semi-autonomous operation. (Localization is provided by a motion capture camera system.) If onboard state estimation via cameras is needed, then a 72 g Odroid-xu3 microcomputer can be used instead [2]. Current ongoing work in our group demonstrates that fully autonomous operation (including localization and mapping) can be performed by employing a 450 g (including the case, SSD, and RAM) Intel NUC mini PC (<http://www.intel.com/content/www/us/en/nuc/nuc-kit-nuc5i7ryh.html>). As a matter of fact, control electronics are getting less expensive, more powerful, and with smaller footprints, following trends predicted by Moore's law [81]. One would argue that the same holds also for light-weight sensors,³ thus constantly driving the weight of an aerial vehicle down.

Unfortunately, batteries do not follow Moore's law, and continue to account for 25–50% of the weight of small UAVs [18]. Therefore, further optimization is required. To this end, current aerial robot designs can benefit from following closer existing aerodynamic and light-weight structure design principles for efficiency [79]. An alternative way is to reconsider the vehicle design in its entirety. The latter is the focus of the section that follows.

2.2. Improving efficiency via mechanical (re-)design

Another hardware-based means to increase the endurance of an aerial robotic vehicle is by making appropriate modifications in its mechanical design. This is a less explored area than weight reduction, yet some interesting approaches have been proposed.

Driessens & Pounds [59] introduce a more efficient rotor configuration termed 'triangular quadrotor'. The proposed configuration employs a centrally located large rotor tasked to generate lift, and three smaller peripherally arranged rotors that are used for control. The small rotors are tilted so to produce the appropriate counter-moment to the vehicle. This arrangement aims to combine the energetic efficiency of a helicopter with the mechanical simplicity of a traditional quadrotor. Compared with a traditional quadrotor, the proposed triangular quadrotor is found to achieve a 15% reduction in the power required to hover [59].

Another interesting approach adds servo motors to the tip of each arm of a quadrotor design to allow tilting of the vehicle's motors [60]. This gives rise to an overactuated system. While this type of overactuation does not seem to offer any benefits in terms of energetic efficiency in hover, it may show promise during forward flight. Indeed, it is plausible that overactuation may actually help design more energy-efficient trajectories to be tracked by the vehicle in forward flight.⁴

2.3. Discussion

Overall, when designing energy-efficient aerial vehicles it is important to take into consideration their intended operation and the associated task specifications (e.g. payload capacity, autonomy capabilities, desired endurance, etc.). To this end, there is need for careful component selection, which turns out to be an interesting system design problem. Critical in this process is the selection of the propulsion system, which also consumes most of the power [43]. For robots capable of hovering, the rule of thumb is that components should be selected so that the vehicle's thrust-to-weight ratio is about two during hover, and while carrying the desired

payload. This way, the vehicle will be responsive to user commands, and able to operate smoothly without risking motor saturation during a possible aggressive manoeuvre. For rotorcraft, in particular, the diameter and pitch of the propeller are important determinants of energy efficiency. Indeed, as the radius of the propeller r_p increases, the rotor produces more thrust T_r per unit power P_r as suggested by the simplified expression in hover [65],⁵

$$P_r = \frac{T_r}{\eta_r r_p \sqrt{2\rho_a \pi}}. \quad (2.1)$$

In (2.1), ρ_a is the density of air and η_r is the figure of merit for the particular system—this is a measure of the efficiency of the rotor [65]. However, merely using larger propellers does not solve the problem because larger propellers require a larger frame to fit, which in turn increases the weight of the vehicle. Similarly, the rotor efficiency increases when one uses high voltage batteries, together with motors of low motor velocity constant (K_v is expressed in r.p.m. per volt), and long propellers. But the weight of the vehicle also increases.

Remark: We emphasize that this is an intentionally brief discussion on component selection, offered here in an effort to give the reader a flavour of the delicate trade-offs when designing an aerial vehicle. Tools like ecalc (<http://www.ecalc.ch/>) offer a place to start at, but actual testing is in fact needed to properly design an aerial system according to given specifications.

The vehicle's endurance t_e (in hover) can then be estimated as the ratio of the total energy available to required power,

$$t_e = \frac{E_b m_b}{n_p P_r + P_i}. \quad (2.2)$$

In (2.2), E_b is the battery specific energy (in Wh kg⁻¹), m_b is the mass of the battery in kilograms, n_p denotes the number of rotors and P_i represents the power required by all other components (i.e. avionics and sensors). Abdilla *et al.* [82] develop a more detailed model to estimate the endurance in hover.

While less explored, thinking out of the box during initial mechanical design may have a large impact in terms of energy efficiency. Key to this is a solid understanding of how thrust is being generated for a particular propulsion system. For example, the triangular quadrotor [59] was motivated by the observation that the power required to hover is inversely proportional to the propeller radius (see (2.1)). Given a vehicle's footprint, a single large rotor will then outperform (energetically) several smaller rotors. The smaller rotors are still needed though to improve the controllability properties of the system [59].

In all, trying to keep the weight as low as possible is guaranteed to increase the endurance of the vehicle. There exist cases, however, where this is not possible. For instance, one needs to work with a preconfigured system, or the system is already well configured. In such cases, algorithm-based solutions can be brought to bear on sustaining long-term operation. We address this topic next.

3. Algorithm-based efficiency optimization

In this section, we discuss how to save energy and extend endurance for robotic flight via software. We present recent

results in energy-aware motion planning for aerial vehicles—rotorcrafts in particular. We also discuss approaches to capture and model various aerodynamic effects that can be incorporated into motion planners and controllers to reduce energy expenditure. The various aerodynamic effects we consider typically depend on the vehicle's ground speed. To this end, we also report some preliminary new results on energy consumption of high-speed robotic flight, indoors.

3.1. Energy-aware motion planning

The problem of generating energy-optimal paths for a rotorcraft has only recently started receiving attention. One idea is to formulate an optimal control problem to generate paths that minimize the energy consumed with respect to motor angular velocities and accelerations [61], or only motor angular velocities [60]. For a traditional quadrotor configuration, and under the condition that the motor angular velocities are identical at the initial time, t_i , and final time, t_f , a candidate cost functional is [61]

$$E = \int_{t_i}^{t_f} \sum_{j=1}^4 (c_0 + c_1 \omega_j(t) + c_2 \dot{\omega}_j^2(t) + c_3 \omega_j^3(t) + c_4 \dot{\omega}_j^4(t) + c_5 \dot{\omega}_j^2(t)) dt. \quad (3.1)$$

The constants c_i , $i = 1, \dots, 5$ are determined based on the parameters of the motors, and on the geometry of the propellers. ω_j and $\dot{\omega}_j$, $j = \{1, 2, 3, 4\}$ denote the j th motor angular velocities and accelerations, respectively. The optimal control problem is further simplified via an appropriate change of variables and then solved numerically; the reported approach is tested in simulation and shows promising results [61]. The reported analysis can perhaps be adapted to other types of multicopter vehicles.

For the overactuated quadrotor configuration [60], a different cost functional based on motor angular velocities only is proposed. However, it appears that the reported cost functional does not fully exploit the overactuation feature of the employed vehicle. The vehicle is also found less energy-efficient in hovering compared with the traditional quadrotor configuration [60]. Still, appropriate shaping of the employed cost functional may yield improved energy efficiency in forward flight.

A different approach focuses on coverage path planning with energy-based optimization criteria [62]. Each path is designed as a concatenation of elementary manoeuvres, with predetermined energy consumption. These include the energy consumed for take off (E_{toff}) and landing (E_{land}) with respect to a fixed setpoint, turning in place (E_{turn}) at a constant yaw rate, and moving on a straight line in two dimensions at constant velocity (E_{move}). Straight-line segments comprise a smooth acceleration phase, a constant-velocity part, and a smooth deceleration phase. The energy consumption of each of these segments is also predetermined, and added up to yield E_{move} . In this light, the total energy consumption for a path consisting of n_{sl} straight-line segments, and n_t turns can be calculated by [62]

$$E_{\text{total}} = E_{\text{toff}} + E_{\text{land}} + n_t E_{\text{turn}} + \sum_{n_{\text{sl}}} E_{\text{move}}. \quad (3.2)$$

Then, an optimization routine identifies the optimal forward velocity in the constant-velocity part so that the energy cost (3.1) is minimized.

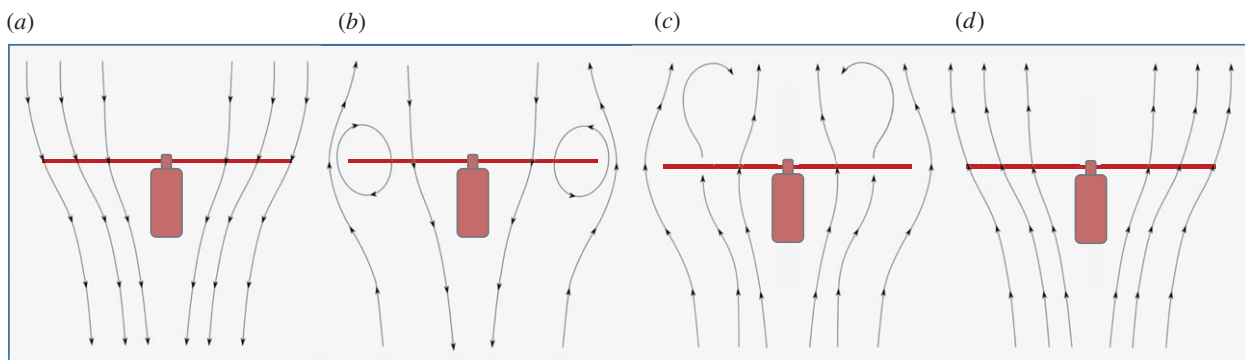


Figure 3. Illustration of the four states that a rotor undergoes during rapid vertical descent. (a) Normal state, (b) Vortex Ring state, (c) Turbulent Wake state and (d) Windmill Brake state. In all cases, arrows show the direction of the flow through the rotor plane. (Online version in colour.)

Taking into consideration the effect of aerodynamics can further improve energy-aware planning algorithms. One example in this realm focuses on path planning for multi-target missions using a hexacopter aerial vehicle [63]. The goal is to identify paths of minimum length while minimizing energy expenditure to move along a sequence of position and yaw waypoints. The reported approach employs a dynamic model that incorporates some aerodynamic effects, and considers various distance heuristics to approximate energy consumption. Simulations indicate that energy-based optimization helps derive tours between the assigned targets that are up to 11.4% more energy-efficient compared with tours that do not account for energy expenditure [63].

Moving a step forward, Ware & Roy [64] integrate an aerodynamics-based model for power consumption in rotorcraft forward flight [65,67] with empirical observations. The derived model is subsequently employed for finding minimum-energy trajectories for motion planning in a wind field. The power per rotor P_r required in forward flight is related to the produced thrust T_r by [65]

$$P_r = \frac{T_r(v_i + v_\infty \sin \alpha)}{\eta_r}, \quad (3.3)$$

where α is the angle of attack, v_∞ is the free-stream velocity and v_i is the induced velocity. The induced velocity is the rate at which the motion of the rotor moves air perpendicular to the rotor plane. It can be calculated by solving the quartic equation [65]

$$v_i = \frac{v_h^2}{\sqrt{(v_\infty \cos \alpha)^2 + (v_i + v_\infty \sin \alpha)^2}}. \quad (3.4)$$

In (3.4), v_h is the induced velocity at hover, given by

$$v_h = \sqrt{\frac{T_h}{2\rho_a \pi r_p^2}}, \quad (3.5)$$

with T_h being the required thrust to hover, that is $T_h = mg/4$ for a quadrotor of mass m . Note that in the case of hovering without any wind effects $v_\infty = 0$, $T_h = T_r$ and (3.3) reduces to (2.1).

Constraining the vehicle to operate at a fixed height and fixed ground speed v_g , the energy consumed for moving from state q_i to state q_j can be then calculated through [64]

$$E_{\text{segment}} = \frac{4P_r(v_g - v_w)\|q_j - q_i\|}{v_g}, \quad (3.6)$$

where v_w is the wind speed and $\|\cdot\|$ denotes the Euclidean norm. Adding the energy consumed to perform consecutive

straight-line segments—that form a desired path—yields the total energy consumed for a specific path. With this information at hand, the derived wind-aware motion planner [64] is found superior to a naive planner unaware of aerodynamics, especially at wind speeds over 10 m s^{-1} .

The aforementioned works [63,64] highlight the significance of including aerodynamic effects in motion planning and control algorithms when considering energy-aware approaches. Providing compelling answers to this problem can significantly improve the performance of aerial vehicles by exploiting aerodynamics, such as the effect of ambient wind [64]. To this end, the question of how to model aerodynamic effects naturally arises.

3.2. Modelling aerodynamic effects

We focus here on three main aerodynamic effects for rotorcraft vehicles: high-speed vertical descent, drag-like effects and the ground effect.

3.2.1. Effect of vertical descent

First, consider vertical descent at a climb velocity v_c . Then, four flight states can be defined [66] with respect to the induced velocity at hover, v_h .

- Normal state when $-1/2 < v_c/v_h$; the air through the rotor is flowing down.
- Vortex Ring state when $-1 < v_c/v_h \leq -1/2$; the air flow starts recirculating, the rotor efficiency drops, and more power needs to be supplied to exit this state.
- Turbulent Wake state when $-2 < v_c/v_h \leq -1$; the air flow recirculation becomes more random, the rotor efficiency significantly drops, and even more power needs to be supplied to exit this state.
- Windmill Brake state when $v_c/v_h \leq -2$; the air through the rotor is flowing up, transferring power to the air.

Momentum theory can be used to provide models for the Normal and Windmill Brake state [83]. However, momentum theory fails in the Vortex Ring and Turbulent Wake states. In these states, the induced velocity can be estimated through an empirical model that is a quartic function of v_c/v_h [65]. Figure 3 shows pictorially the four states at vertical descent for the case of a single rotor. Overall, if rapid descent needs to be performed, it is recommended to allow for some small forward velocity component, if possible, in order to reduce efficiency losses [67].

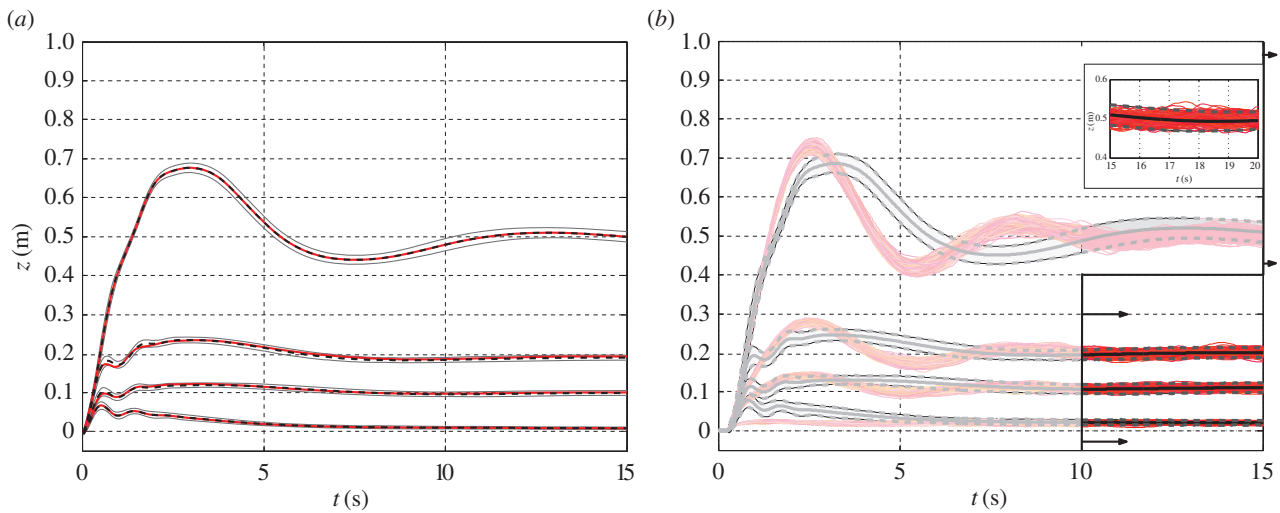


Figure 4. (a) Data-driven model performance using the POD-based approach [68]. Model-predicted response (shown in red) is plotted against sample means (dashed black curves) and one standard deviation around the latter (solid grey curves). The predictive capacity of the derived data-driven model is tested and validated against the system behaviour when tasked to hover at $|z| = 20$ cm from the ground. The graph is taken from Karydis & Hsieh [68] with permission from the authors. (b) Performance of the stochastically extended model [69] to capture the steady-state behaviour of a quadrotor tasked to hover at various heights from the ground. Monte Carlo simulation of the derived model reveals that the latter can capture the variability observed in experimental data. Red curves indicate individual stochastic model realizations, and are plotted against the sample means (solid thick curves) and three standard deviations (dashed thick curves) of experimental data. The graph is taken from Karydis *et al.* [69] with permission from the authors.

3.2.2. Drag effects

We now turn our attention to drag and drag-like effects. Bangura & Mahony [66] propose a simple lumped parameter nonlinear model for drag effects. The model can be incorporated in the rotorcraft dynamics, and it is designed so that it is valid over the full vehicle flight envelope (not just in steady state). Five drag components are considered.⁶

- Blade flapping, which is proportional to the vehicle's angular velocity and linear velocity in the x - y plane.
- Induced drag, which is proportional to the vehicle's planar linear velocity.
- Parasitic drag, also proportional to planar linear velocity.
- Translational drag, which is proportional to planar linear velocity but at high velocities the vertical component of the vehicle's velocity and the induced velocity need to be accounted for.
- Parasitic drag, which scales with the square of the three-dimensional velocity of the vehicle.

The total drag force acting on the vehicle is the sum of all the above components, taken over all rotors. The first four drag components also create additional torque. The torques created as a result of drag forces on individual rotors are added to take the total additional torque on the vehicle [66]. Including the effect of drag as additional force and torque terms in a model for a rotorcraft is helpful in two ways. Foremost, it allows the designer to study how the energy expenditure due to drag varies. Second, it enables the generation of motion trajectories that explicitly take into consideration the effect of drag. Similar benefits can also be gained by modelling ground effects.

3.2.3. Ground effect

The ground effect is known to manifest itself through an increase in the lift of the vehicle. Specifically, the downward airflow generated by the rotors of the platform redirects the flow parallel to the ground plane, which in turn increases

the effective lift generated by the system. Blade element and momentum theory has been used to provide an analytical model to capture the increase in the effective lift for a single rotor as [84]

$$\frac{T'_r}{T_r} = \frac{1}{1 - (r_p/4|z|)^2 \{1/1 + (|V|/v_i)^2\}}. \quad (3.7)$$

In (3.7), r_p is the radius of the propeller, $|z|$ is the distance from the ground, $|V|$ is the vehicle's speed, and v_i is the induced velocity. T'_r and T_r denote the thrust close to and away from the ground, respectively. Note that (3.7) indicates that the ground effect practically diminishes when $|z|/r_p \geq 2.5$. However, this result is for one rotor, and does not take into consideration the effect of adjacent rotors in a multi-rotor vehicle. This perhaps explains the discrepancies observed between (3.7) and the experimental results obtained in some recent studies on ground effect for rotorcraft [85–87].

The ground effect can also be captured through data-driven [68] or appropriate low-dimensional stochastic [69] models. Karydis & Hsieh [68] derive a data-driven model based on the principal orthogonal decomposition (POD) [88,89] to capture the ground effect for a hovering quadrotor. The idea is that primary modes extracted from altitude data away from the ground capture the principal dynamics (in this case, of a second-order system). The principal mode is scaled and subtracted from data collected at other heights inside the ground effect zone. Using POD again, secondary modes are subsequently extracted. The latter are also scaled and subtracted from data of hovering very low to the ground. The primary mode of what remains is termed the 'ground effect mode'. It is found that three modes (ground effect, principal and an intermediate one) suffice to predict the average behaviour of the system in other hovering altitudes [68]. This is achieved by identifying appropriate scaling functions that can map the constructed principal modes back to altitude data for hovering at different setpoints. Figure 4a demonstrates this point pictorially. The proposed model is trained based on the datasets collected when the vehicle was

commanded to hover at 50 cm, 11 cm and 2 cm, and tested against the dataset with the hovering setpoint at 20 cm. With reference to figure 4a, sample means are marked with dashed black curves and one standard deviations around the mean are shown with solid grey curves. The output of the proposed data-driven model based on POD analysis closely predicts the sample mean of the testing dataset at 20 cm, and even captures the transient effects.

In a separate effort, Karydis *et al.* [69] propose a systematic approach to bridge data-driven and model-based machine learning. The reported approach starts with a deterministic model that captures salient system behaviours, and extends it to a stochastic regime by turning its parameters to random variables. Experimental data are then used to estimate the parameters of the distribution that the random variables should be drawn from. The parameter estimation is performed so that the output of the derived stochastic model captures the variability observed in the original system behaviour. The approach comes with probabilistic guarantees on the validity of the outcome [69]. Figure 4b shows its application on the same dataset as before [68]. The model tested is a double integrator with zero-mean Gaussian noise corrupting its output. The variance of the noise is identified in all cases so that the output of the stochastic model captures the mean and variance of the experimental data at all setpoints in steady state.

These techniques can be used, in principle, to capture ceiling effects. The ceiling effect is in some sense the opposite of the ground effect: the pressure above the vehicle gets lower, and the vehicle gets pulled toward the ceiling.

3.3. High-speed flight

The aforementioned results highlight that power consumption and the ‘strength’ of aerodynamic effects are related to the ground speed of the robot. To calculate the effect of increased ground speeds on power consumption, we collected data from an Asctec Hummingbird quadrotor of mass 0.551 g flying in straight-line, constant-velocity trajectories, indoors. The commanded speeds varied in the interval $[0, 8] \text{ m s}^{-1}$.⁷ Position, velocity and orientation data were collected with a motion capture camera system at 100 Hz. We also collected IMU measurements and commanded thrust at 100 Hz. From (3.3), (3.4) and (3.5), we calculated the ratio of the power consumed at forward flight, P , to the power consumed at hover, P_h . The results are shown in figure 5 as a function of ground speed. The calculated values for the force ratio are marked with crosses. We observe that the power consumption steadily decreases for speeds in the range $1.5\text{--}6 \text{ m s}^{-1}$, and then starts increasing. This reduction in power consumption can be associated with translational lift; more air flows through the rotors in forward flight, thus improving rotor efficiency [65].

The data are fitted well with a third-order polynomial ($R^2 = 0.9936$); this is shown in figure 5 with a solid curve. Using this third-order polynomial model, we attempt to predict the power consumption beyond 8 m s^{-1} ground speed where no data are available. The prediction is marked with a dashed curve in figure 5. According to this prediction, the power consumption increases steadily beyond 8 m s^{-1} , reaching the power consumption at hover at about 9.5 m s^{-1} .

Our results in the region $0\text{--}8 \text{ m s}^{-1}$ agree qualitatively with those reported by Ware & Roy [64, fig. 4b]. Note that

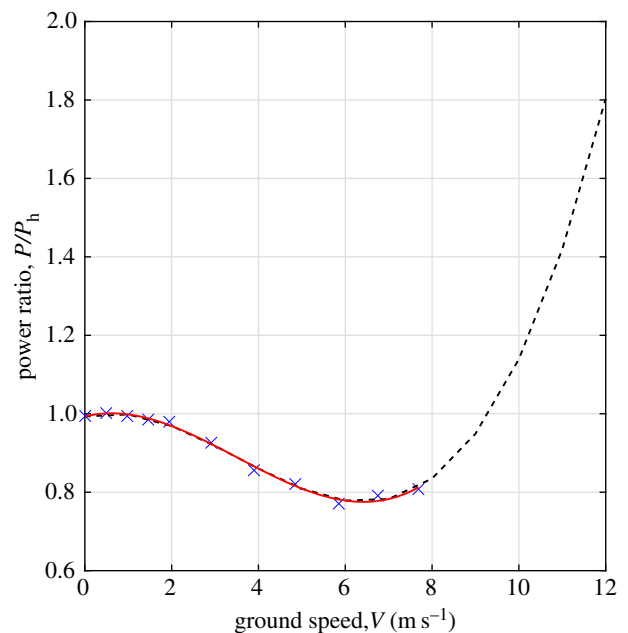


Figure 5. Power consumption data normalized by the power consumption at hover. Blue markings denote the results calculated by (3.3)–(3.5) using the measured velocity from motion capture, and the thrust commands sent to the vehicle. A third-order polynomial model fits the data well (shown with a thick curve), while its prediction at speeds over 8 m s^{-1} (shown with a dashed curve) seems to be in agreement with the theory [66].

results agree despite that data in the two studies were collected with different quadrotors, and at different settings.⁸ Quantitatively, our data show lower power consumption than [64]. This mismatch may be caused by differences in the drag affecting the two vehicles, or by discrepancies in the pitch of the vehicle due to the distinct experimental settings employed at the two studies. As a final remark, it appears that the predicted power consumption agrees qualitatively with the analysis of drag effects. The analysis suggests that the parasitic drag (which scales with the square of the velocity) becomes important at ground speeds over 10 m s^{-1} [66]. Still, more data at higher speeds are needed to test the hypothesis that the identified third-order polynomial model is actually valid at higher speeds.

3.4. Discussion

Modelling the various aerodynamic effects is important as they are ultimately linked to energy losses (as in the cases of vertical descent and drag effects) or gains (as in the case of ground effects). At the same time, incorporating the derived models of aerodynamic effects in motion planning and control algorithms leads to more accurate energy-aware algorithms. In cases where no further hardware optimization is possible, algorithm-based approaches can still drive the energetic cost of robotic flight down.

The motion planning approaches reported here employ a feedforward control paradigm to handle aerodynamic effects. Closing the loop around energy-aware controllers has received less attention. One approach is to consider the model of the system as a hybrid one [66]. The states of the model depend on the operation conditions—for example in ground effect zone, or at Vortex Ring state—and a hybrid model predictive control scheme [90] can be closed around

this model. The need for energy-aware feedback controllers becomes even more pressing if one considers the rapidly increasing number of multimodal aerial robots that could soon outperform traditional monolithic aerial robots.

4. Multimodal operation

Aerial robots—especially those that hover—are by their very nature inefficient because they consume a substantial amount of power to sustain lift (approximately 200 W kg^{-1} [18,43]). The way to mitigate this problem is by considering multimodal aerial robots. Inspired by nature, multimodal aerial robots is a nascent but rapidly growing area of research. In this section, we examine two categories: (i) aerial robots capable of perching and (ii) aerial robots able to simultaneously walk or roll.

4.1. Perching

Several works have focused on endowing UAVs with perching capabilities. The rationale is that when perched, an aerial robot would minimize its energy expenditure but could still be functional [2]. For example, a perched robot could provide situational awareness or monitor a site from atop. Successful perching on vertical walls has been achieved using prongs (or spines) that penetrate the surface with a micro-glider [91] and a fixed-wing vehicle [92]. The insect-scale flapping-wing robobee [6] successfully perched on vertical surface using magnetic adhesion [71], and recently on a variety of overhanging surfaces using electrostatic adhesion [72].

The perching capacity of rotorcraft, and quadrotors primarily, is growing fast. Gecko-inspired adhesives have been found successful for perching on smooth inclined and vertical surfaces [70,93,94]. In addition, rotorcraft perching can be achieved via various attachments such as suction-based mechanisms [95], spines [96], and passive [97,98] and active [2,99] grippers.

4.2. Walking and rolling

Less explored in the literature is the energetics of aerial vehicles that can also walk (like birds) or roll. Recent results consider centimetre-scale quadrotors with the capacity to either walk with legs attached below them [73], or roll while being suitably hinged inside a cylindrical cage [75]. The latter configuration enables the robot to increase its operation range compared with the flying-only case. Energy savings during terrestrial locomotion when compared with aerial locomotion are studied and experimentally validated [75]. Depending on the surface, the robot's terrestrial range and operational time are found up to 11 and 10 times greater than the range and operation time at equivalent speeds while flying, respectively [75].

A different approach considers a light-weight bipedal ornithopter that can run fast and transition between aerial and terrestrial locomotion modes [35]. The flapping wings of the robot are found to offer damping, propulsive force, and to contribute to its dynamic stability; the robot can run bipedally with only a single actuator. To transition from ground running to aerial hovering, the robot requires about 1 m of runway [35]. Long distance flight and terrestrial locomotion in cluttered environments can be achieved through a bi-modal morphing-wing robot [74]. The wing tips can

morph into rotary legs upon landing, to facilitate local exploration and increase the robot's efficiency. Aerial and terrestrial capabilities are powered by a single locomotor apparatus, thus reducing the total complexity and weight of the robot [74].

4.3. Discussion

Depending on the application, robotic vehicles capable of multimodal locomotion can be more efficient than their flying-only counterparts. However, more studies are needed to better understand multimodal vehicle efficiency. Then, algorithmic optimization of flying and walking behaviours will be enabled. Most recent works have focused on vehicle design, and have considered perching and capability for terrestrial locomotion separately. Yet, it may make sense to eventually integrate these two functions together. In all, recent results seem to confirm that multimodal vehicles offer tremendous potential.

5. Conclusion

We discussed the development in aerial robotics from the point of view of energetics. To improve energy efficiency, we can look at three critical aspects. The first aspect centres on careful component selection and design optimization. The second one concerns clever algorithmic design of motion planners and controllers that are energy-aware and manage to harness various aerodynamic effects. The third aspect is through multimodal locomotion. We also presented recent efforts on modelling the effect of aerodynamics, and integrating aerodynamic cost functions to optimize flight paths and behaviours, including high-speed flight. Improving the energetics of flight is key to realizing the full potential of aerial robots in many real-world applications.

To further improve the energetics of robotic flight, we need to combine expertise from several fields. These include robotics, fluid mechanics, material sciences and biology of avian flight, among others. Several works presented herein are along these lines. But more interdisciplinary, collaborative efforts are needed. We hope that this review paper will motivate more collaborations among the different fields related to the energetics in robotic flight. Conversely, birds can lower the energetic demand of flight by reducing the amount of flapping, or flying in formation [100]. What if additional ways to reduce the cost of flight can be revealed via robotic path optimization based on aerodynamic cost functions? This way the methods described in §3 could feed back to biology by lending themselves as new tools to study the energetics of animal flight.

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Endnotes

¹Tethered-based solutions have started gaining momentum, and the control theory to support this area is also being developed [52].

²For example, the Intel NUC mini PC: <http://www.intel.com/content/www/us/en/nuc/nuc-kit-nuc5i7ryh.html>.

³Consider for instance recent developments in LiDARs, which are critical for mapping, localization and autonomous navigation and obstacle avoidance.

⁴More about energy-aware trajectory generation follows in §3.

⁵A list of symbols used in this review is provided in appendix A.

⁶The interested reader is referred to [66] for the detailed analysis and derivation of the expressions for the various drag components.

⁷Owing to space restrictions, we were not able to fly trajectories at speeds higher than 8 m s^{-1} .

⁸Compare flight in place inside a wind tunnel [64] with forward flight indoors.

Appendix A. List of symbols

For the convenience of the reader, we present here a collection of the key symbols used in this review.

c_i	i th optimization constant
m_b	battery mass
n_p	number of rotors
n_{sl}	number of straight-line segments in a trajectory
n_t	number of turns in a trajectory
q_i	i th state in waypoint-based trajectory
r_p	propeller radius
t	time

t_e	vehicle endurance
t_f	final time for optimization
t_i	initial time for optimization
v_∞	free-stream velocity
v_c	climb velocity
v_g	ground speed (equal to airspeed indoors)
v_h	rotor induced velocity at hover
v_i	rotor induced velocity
v_w	wind speed
z	vertical distance from the ground
E	generic symbol for energy consumption
E_b	battery specific energy
E_{land}	energy consumption at landing
E_{move}	energy consumption in straight-line segments with smooth acceleration and deceleration
$E_{segment}$	energy consumption in general straight-line segments
E_{toff}	energy consumption at takeoff
E_{total}	total energy consumption of a trajectory
E_{turn}	energy consumption during turns
K_v	motor velocity constant
P_h	power required by rotor at hover
P_i	power required by all non-propulsion systems
P_r	power required by rotor
T_h	thrust produced by rotor at hover
T_r	thrust produced by rotor
T'_r	thrust produced by rotor in ground effect
V	speed
α	rotor angle of attack
η_r	rotor figure of merit
ρ_a	air density
ω_j	angular velocity of the j th rotor
$\dot{\omega}_j$	angular acceleration of the j th rotor

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